



Teheran University of Medical Sciences

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Final Exam
International Campus
TISSUE- DEVELOPMENT
September 2024 semester

Name and Surname:

Student ID Number:

Course Coordinator: Dr. Rastegar

Date of Exam: Jan. 20, 2025

Duration of Exam: 80 minutes

MCQ Question	80
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Directions:

- Each of the following questions contains one correct answer only.
- Please choose the correct answer for the MCQs in the answer sheet with pencil.



Histology: Dr. Rastegar

- 1- Most important function of which tissue is heat production?
a) Tendon b) Areolar tissue c) brown Adipose tissue d) elastic tissue
2. Which one is dense regular connective tissue?
a) Dermis of skin b) Tendon c) Adipose tissue d) Areolar tissue
3. In which cell we have major basic protein?
a) lymphocyte b) monocyte c) macrophage d) Eosinophil
4. Which cell has most number in the normal blood smear?
a) Neutrophil b) Basophil c) monocyte d) Eosinophil
5. Which cell has no nucleus?
a) plasmacell b) lymphocyte c) red blood cell d) monocyte
6. Russel bodies are characteristic for which cell?
a) mastcell b) plasmacell c) kupfer d) Adipocyte
7. structure of framework of spleen and lymphnodes are composed of which fibers?
a) reticular b) elastic c) elaunin d) collagen bundles
8. which one is dense irregular connective tissue?
a) Dermis of skin b) Tendon c) Adipose tissue d) Areolar tissue
- 9- What do you call the simple squamous epithelium that lines the blood vessels?
a. Epithelioid tissue b. Mesothelium
c. Endothelium d. Pseudostratified
- 10 -Which type of junctions included Cadherins?
a. Gap junctions b. Desmosomes
c. Hemidesmosomes d. Tight junctions
- 11- An exocrine gland that loses cell bodies during secretion, is further classified as a _____ gland that Occured in the _____.
a. Merocrine- sebaceous gland b. Holocrine- sebaceous gland
c. Ecrine- Sweat glands d. Apocrine - Sweat glands

12- Identify the INCORRECT statement

- a. The basement membrane consists of two sub layers: the reticular layer and the basal lamina.
- b. Epithelia are classified by the shape of cell layers.
- c. Microvilli increase the surface area of the epithelia cells.
- d. In pseudostratified epithelia all cells are in contact with the basement membrane.

13- Which of this cell included Perivascular feet?

- a. Oligodendrocyte
- b. Astrocyte
- c. Satellite cell
- d. Schwann cell

14 - Each peripheral nerve is surrounded by:

- a. Perineurium
- b. Endoneurium
- c. Epineurium
- d. Myelin sheat

15- Dendrites have all organelles except:

- a. Mitochondria
- b. Lysosomes
- c. Golgi complex
- d. Ribosomes

16- Retrograde transportation is occurred by:

- a. Kinesin
- b. Dynein
- c. Actin
- d. Integrin

17- Which of these is abundant in axon terminals?

- a. Ribosome
- b. Lysosome
- c. Golgi complex
- d. Vesicles

18- Nissl bodies in the perikaryon of neurons are composed of:

- a. Ribosomes and Lysosomes
- b. Ribosomes and RER
- c. Golgi and RER
- d. SER and Lysosomes

19- Spinal central canal covered by:

- a. Ependymal cell
- b. Oligodendrocyte
- c. Schawann cell
- d. Microglia

20- Which types of these neurons are located in motor neurons?

- a. Multipolar
- b. Anaxonic
- c. Bipolar
- d. Pseudounipolar

21- What is the structure of TRIAD?

- a. 2 SER+1TT
- b. 1 SER+2TT
- c. 1 SER+1TT
- d. 2 SER+2TT

22- Which of muscle protein has three subunit(TnT,TnC,TnI)?

- a. Actin
- b. Myosin
- c. Troponin
- d. Tropomyosin



- 23- In which muscle type(s) nucleus located in center part?
- smooth muscle and striated muscle
 - skeletal muscle and cardiac muscle
 - smooth muscle and skeletal muscle
 - cardiac muscle and smooth muscle
- 24- What is Junctional Folding's in muscle tissue?
- Sarcomere
 - Intercalated disc
 - Terminal Axon
 - Muscle cell membrane
- 25- Which layer of the epidermis is present only in thick skin?
- Stratum basale
 - Stratum spinosum
 - Stratum lucidum
 - Stratum corneum
- 26- Which layer of the epidermis has cells which have keratohyaline granules?
- Stratum basale
 - Stratum spinosum
 - Stratum granulosum
 - Stratum corneum
- 27- What type of tissue makes up the dermis of the skin?
- Mucus
 - Mesenchyme
 - Muscular
 - Connective tissue
- 28- Which cell is a macrophage found in the skin?
- Langerhans cell
 - Dust cell
 - Microglia
 - Histiocyte
- 29- Mesenchymal Cell Like Is Observed-----?
- In surface layer of perichondrium
 - In chondrim
 - In middle layer of perichondrium
 - In Lacuna
- 30- Which part of the following is observed collagen Fiber II?
- Elastic cartilage tissue
 - Hyalin cartilage tissue
 - Fibro cartilage tissue
 - Perichondrim tissue
- 31- In microscopic structural of joint tissue which part of the following is observed F Cell & M Cell?
- End plate bone
 - Synovial membrane
 - Synovial cavity
 - Synovial Fluid

2. In microscopic structure which part of the Diarthrosis joint is observed Hyaline cartilage tissue without perichondrium tissue?

- A. Articular cartilage Tissues
- B. End plate bone
- C. Lamina propria synovial membrane
- D. Nothing

33. Which parts of the following is observed osteogenic cells?

- A. Outer layer of periosteum
- B. Osteoid tissue
- C. Inner Layer Of Periosteum
- D. Lacuna

34. Which part of the following is observed Endosteum tissue?

- A. Metaphysis plate tissue
- B. Central canal Haversian system Tissue
- C. Immature bone tissue
- D. Outer peripheral bone tissue

35. Which of the following is observed Irregular Haversian System?

- A. Compact bone tissue
- B. Alveolar Bone Tissue
- C. Mature bone tissue
- D. Spongy bone tissue

36. Osteoclast is observed?

- A. In Endosteum Tissue
- B. Around lacuna
- C. Between Haversian system
- D. In periosteum tissue

Embryology: Dr. Amidi

37- The following transformations of the spermatid take place in spermiogenesis (stage of formation), EXCEPT:

- A - the nucleus becomes decondensed;
- B - Golgi complex produces an acrosomal granule that is converted to an acrosome;
- C - mitochondria form the helical sheath round the axonemal complex;
- D - the distal centriole initiates formation of the tail

38- Each of the following statements concerning the ovum penetration by a sperm is true, EXCEPT:

- A - acrosomal enzymes facilitate the sperm passage through the oocyte membrane
- B - the oocyte and sperm plasma membranes fuse and break down at the point of contact;
- C - the sperm plasma membrane enters the oocyte cytoplasm;
- D - the sperm nucleus enters the oocyte cytoplasm.



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39- Each of the following statements concerning the human cleavage is true, EXCEPT:

- A - is a process of successive rapid mitotic divisions;
- B - blastomeres increase in volume becoming progressively larger;
- C - proceeds in the oviduct during the first three days when the morula is formed;
- D - continues in the uterus during 3 or 4 days when the blastula is formed;

40- The following statements regarding the mesoderm differentiation are true: EXCEPT

- a) paraxial mesoderm divides into somites
- b- intermediate mesoderm forms the somites
- c- lateral mesoderm is divided into somatopleure and splanchnopleure
- d- somatopleure and splanchnopleure surround the coelom

41- The events of the second week of embryonic development are as follows, EXCEPT:

- A - blastocyst formation;
- B - the first stage of gastrulation;
- c - chorion development;
- d - yolk sac formation.

42- The placental barrier includes the following structures, EXCEPT:

- A - syncytiotrophoblast of chorionic villi;
- B - connective tissue of the decidua basalis;
- C - cytotrophoblast
- D - connective tissue of tertiary villi;

43- The events of the first week of the human embryonic development are as follows: EXCEPT

- a. fertilization
- b. cleavage
- c. blastula formation
- d. gastrulation

ues except:
d trophoblast

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44- The placenta is the major endocrine gland producing the following hormones:

- a. human chorionic gonadotropin (hCG)
- b. human chorionic somatomammotropin (hCS)
- (c) relaxin
- d.) estrogens and progesterone

45- Oogenesis is arrested in which stage of second meiotic division

- a. Prophase
- b. Metaphase
- c. Anaphase
- d. Telophase

46- After ovulation, the remanent of Graafian follicle is called as

- a. Corpus albicans
- b. Corpus luteum
- c. Corpus atreticum
- d. Corpus hemorrhagicum

47- Which of the following is true about zona reaction?

- a. Release of enzymes from acrosomal cap
- b. Secretion of hyaluronidase enzyme by zona pellucida
- c. Glycoprotein coat of sperm gets dissolved
- d. Zona pellucida becomes impermeable to other sperms after entry of one sperm

48- Roof of secondary yolk sac is formed by

- a. Amnioblast
- b. Hypoblast
- c. Chorion
- d. Trophoblast



49- The week of "two's" is characterized by formation of the following structures except:

- a. Amniotic cavity and yolk sac
- b. Syncytiotrophoblast and cytotrophoblast
- c. Hypoblast and epiblast

d. Amnioblast and trophoblast

50- The process of formation of intraembryonic mesoderm from primitive streak is known as

- a. Blastulation
- b. Neurulation
- c. Gastrulation
- d. Invagination

51 - Nucleus pulposus of the intervertebral

disc is remanent of

- a. Notochord
- b. Primitive streak
- c. Primitive knot
- d. Prechordal plate

52- Maternal component of placenta is formed by

- a. Decidual basalis
- b. Chorion frondosum
- c. Decidua parietalis
- d. Chorion leave

53- Failure of complete division of inner cell mass results in formation of

- a. Conjoined twins
- b. Monozygotic twins
- c. Dizygotic twins
- d. Quadruplets

54- Sacrococcygeal teratoma comes from remnant of;

- a- allantois
- b- somites
- c- notochord
- d- primitive streak

55- Regarding Oogenesis all statement are true, Except:

- a- begins in female before birth & completed after puberty
- b- no primary oocyte form after birth in female
- c- oocyte has the abundance of cytoplasm
- e- usually two ovum mature every month

56- Turner syndrome is due to:

- a- Monosomy of the X chromosome
- b- Trisomy of the 18 chromosomes
- c- Monosomy Y chromosome
- d- Trisomy of the 13 chromosomes

57- During spermatogenesis, chromosome number is halved at the conversion step of

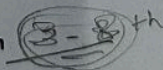
- a. Spermatid to spermatozoon
- b. Spermatogonium to primary spermatid
- c. Primary spermatocyte to secondary spermatocyte
- d. Secondary spermatocyte to spermatid

58- Medulla of adrenal gland develops from

- a. Endoderm
- b. Neural crest
- c. Intermediate mesoderm
- d. Paraxial mesoderm

59- Organogenesis is the period between

- a. The 3rd and 7th weeks X
- b. The 4th and 8th weeks
- c. The 2nd and 5th weeks
- d. The 2nd and 8th weeks





60- Noninvasive prenatal diagnostic technique is

- a. Amniocentesis
- b. Ultrasonography
- c. Chorionic villus sampling
- d. Fetal blood sampling

61- Umbilical cord contains

- a. Two arteries and one vein
- b. Two veins and two arteries
- c. One artery and two veins
- d. Two veins and one artery

62- When a pregnant woman starts to feel their baby's movement in their uterus?

- a. Third month
- b. Fifth month
- c. Sixth month
- d. Seventh month

63- Primitive node (node of Hensen) takes place:

- a. On the anterior (cranial) end of the primitive streak
- b. On the posterior (caudal) end of the primitive streak
- c. In the middle of the primitive streak
- d. Outside of primitive streak

Physiology: Dr. Ranjbaran

64- A single contraction of skeletal muscle is most likely to be terminated by which of the following actions?

- a) Closure of the postsynaptic acetylcholine receptors
- b) Removal of acetylcholine from the neuromuscular junction
- c) Removal of sarcoplasmic Ca
- d) Return of the dihydropyridine receptor to its resting conformation

65- Smooth muscle contraction is terminated by which of the following?

- a) Dephosphorylation of myosin kinase
- b) Dephosphorylation of myosin light chain
- c) Efflux of Ca ions across the plasma membrane
- d) Inhibition of myosin phosphatase

66- Which of the following best describes an attribute of visceral smooth muscle not shared by skeletal muscle?

- a) Contraction is ATP dependent
- b) Contracts in response to stretch
- c) Does not contain actin filaments
- d) High rate of cross-bridge cycling

67- Calmodulin is most closely related, both structurally and functionally, to which of the following proteins?

- a) G-actin
- b) Myosin light chain
- c) Tropomyosin
- d) Troponin C

68- The delayed onset and prolonged duration of smooth muscle contraction compared with skeletal muscle, are all consequences of which of the following?

- a) Greater amount of myosin filaments present in smooth muscle
- b) Higher energy requirement of smooth muscle
- c) Physical arrangement of actin and myosin filaments
- d) Slower cycling rate of the smooth muscle myosin cross-bridges

69- Which of the following decreases in length during the contraction of a skeletal muscle fiber?

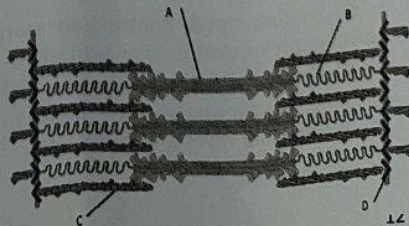
- a) Thick filaments
- b) Thin filaments
- c) Z disks of the sarcomere
- d) Sarcomere

70- Which of the following best describes a physiological difference between the contraction of multi-unit smooth muscle compared with the contraction of skeletal muscle?

- a) Ca independent
- b) Does not require an action potential
- c) Requires more energy
- d) Shorter in duration

71- In the opposite figure, which of the following refers to "Titin"?

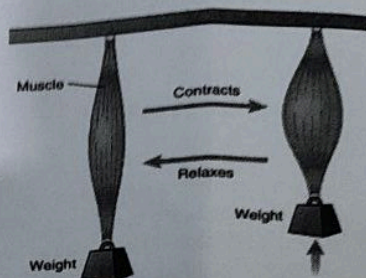
- a) A
- b) B
- c) C
- d) D





72- Which phenomenon is shown in this figure?

- a) The overall length of the muscle does not change.
- b) In this contraction, tension remains constant.
- c) load is greater than the force of the muscle contraction
- d) it shows staircase effect

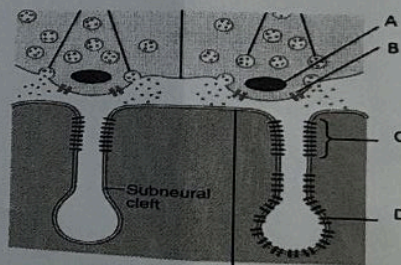


73- Which of the following best describes the significant difference between skeletal muscle and large nerve fibers?

- a) Generation of action potential
- b) Existence of voltage gated sodium channels
- c) Amount of resting membrane potential
- d) Duration of action potential

74- Which of the following represents "voltage gated calcium channels"?

- a) A
- b) B
- c) C
- d) D



75- Caveolae is

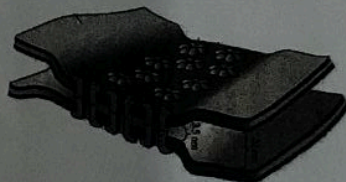
- a) An analog of the transverse tubule
- b) a subunit of troponin complex
- c) active sites of actin filaments
- d) cross bridges of myosin filaments

76- Which of the following best describes the correct temporal order of events for skeletal muscle?

- a) Muscle action potential, Muscle contraction, Nerve action potential
- b) Muscle action potential, Nerve action potential, Muscle contraction
- c) Muscle contraction, Muscle action potential, Nerve action potential
- d) Nerve action potential, Muscle action potential, Muscle contraction

77- Which of the following refers to one property of this type of synapse?

- a) Unidirectional transmission
- b) No synaptic delay
- c) No cytoplasmic continuity
- d) Existence of chemical transmitter



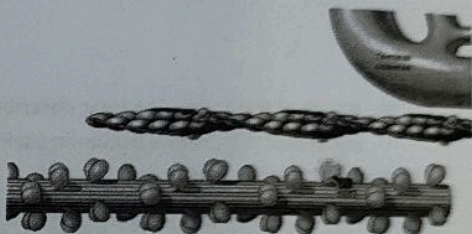


78- During a contraction cycle, what does ATP bind to?

- a) Myosin filaments
- b) Actin filaments
- c) Active sites
- d) Troponin complex

79- What state is shown in the opposite figure?

- a) Relaxation in skeletal muscle fiber
- b) Contraction in skeletal muscle fiber
- c) Relaxation in smooth muscle fiber
- d) Contraction in smooth muscle fiber



80- In this type of muscle, the source of Ca^{2+} ions needed for contraction is just from sarcoplasmic reticulum?

- a) Biceps muscle
- b) Uterus
- c) Iris muscle of the eye
- d) Blood vessels